

Alex Zhang

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Education

- **University of Toronto** | *HBSc, **Physics** Specialist, **Computer Science** Major, **Mathematics** Minor* | 2024 to 2028
 - 3.80/4.00 GPA. Dean's List Honours.
 - Varsity Blues Men's Volleyball practice player.
 - Relevant curric.: Classical Mech. (A+), Intro. Quantum (A+), Intro. Programming & CS (A+, A), ODEs (A), Lin. Alg. I & II (A), Advanced Calc. (A+), Math. Expr. & Rsng. for CS (A+)
- **Merivale High School** | *International Baccalaureate Diploma* | 2021 to 2024
 - 38/45 IB score, Silver Medal, Ontario Scholar, Honours, StudCo., AthlCo., Sr. Boys' Volleyball Captain.

Experience

- **University of Toronto, Department of Physics** | *Research Student* | May 2026 to August 2026
 - PI: Professor Xiang Li.
 - Project: "Characterization of optical properties in titanates using ellipsometry"
- **Ciena** | *Test Engineer Platforms Intern (Vulcan Lab, Product Integration Team)* | May 2025 to August 2025
 - Using **Python, TCP/IP, C#, Linux**, electrical test equipment (oscilloscopes, multimeters, test probes) to configure test cases for **I2C, SPI, JTAG, CRC**-based board functions and device bootup programming.
 - Working with SW & HW Design teams to **troubleshoot, debug** test software and product hardware.
 - Implementing Sequencer test platform & Product Limits database to be **deployed globally** to manufacturers and writing user- & developer-friendly **documentation** on deliverables.
 - **Improved test coverage from 65% to 80%** on Control & Timing Module product board.
- **Leaside Volleyball Club** | *Coach* | December 2024 to April 2025
- **Browns Socialhouse** | *Dishwasher* | June to August 2024
- **HiFresh Foods** | *Warehouse Clerk* | June to August 2024
- **Kumon Kanata North** | *Centre Assistant/Tutor* | July to October 2023

Involvement

- **UT Aerospace Team Space Systems** | *Electrical Team* | January 2025 to April 2025
 - (Student-run club developing FINCH: low-earth orbit CubeSat.)
 - **PCB design** project (see below), assessed **structural integrity** of on-board computer's bus connectors.
- **Arts for Seniors** | *Head of Marketing* | June 2023 to June 2024
 - Managing media and outreach for student-led nonprofit connecting student artists with retirement homes.

Skills/Projects

- **Languages: Python, Java, MATLAB, HTML, CSS**
 - "NBA Guesser", GUI trivia game. **HTML web scraping** to compile data. Thorough documentation.
 - "Pixel Art Canvas", MS Paint-style GUI. Complex class data representation. Multilayered canvas and exportable drawings. Automation of testing.
 - "Raccoon Raiders!", GUI game using recursion to compile and compute requests. Automation of testing.
 - MATLAB Certified
- **PCB Design**
 - Used Altium Designer to create **PCB schematic and layout**, configuring I2C, SPI communication on an STM32 microcontroller-based acceleration sensor board. Datasheet, Schematic **analysis**.
- **Trilingualism (English, DELF B2 certified French, native Mandarin)**, Written communication (Ottawa Library Youth Authors Competition Runner-up)